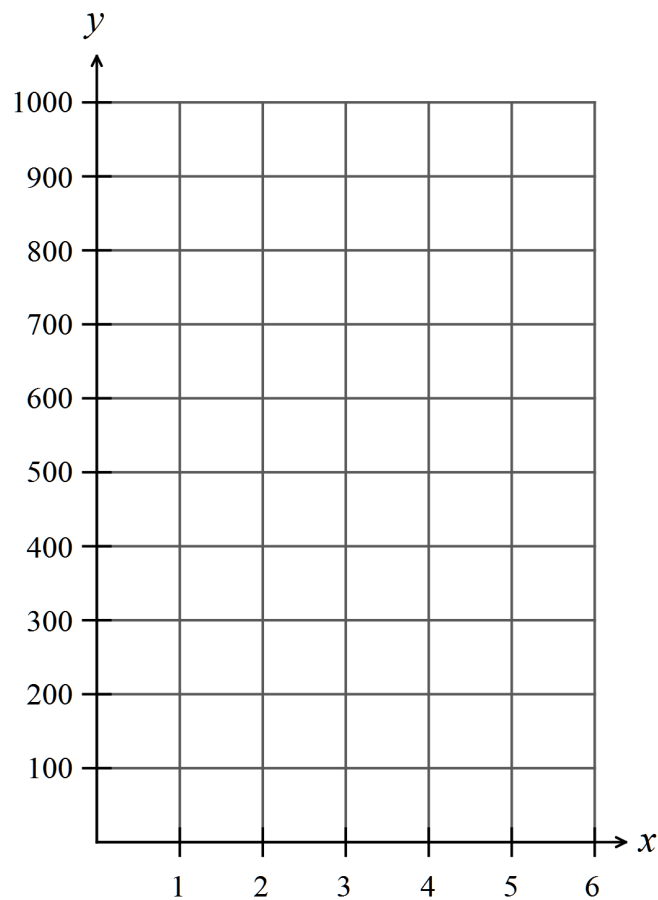


Question 4**(13 marks)**

(a) John had savings of \$200 and planned to save \$150 each week starting next week.

- (i) Determine an equation that gives the amount John had saved in terms of t where t is the number of weeks starting today at $t = 0$. (2)

- (ii) Sketch the equation on the set of axes below. (2)



At the same time, his brother Ian met a girl that intrigued him. He decided to spend \$50 a week on the girl from his savings of \$1000 and get to know her.

- (iii) Determine the equation that represents Ian's savings in terms of t . (2)

